NOISEMAP · DESIGN DECISION RECORD

NoiseMap recommendations accepted

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DOCUMENT CONTROL

Document ID	NSM-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/noisemap		
Source	docs/decisions/0002-recommendations-accepted.md @ ffa26f8		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1 decided by Amish on 2026-10-02 (recommendation approved, NSM-DEC-001)
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and the changes made in this repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below with a recommendation is decided by Amish, 2026-09-25: go with recommendation. O1, which had no recommendation, was decided on 2026-10-02 (NSM-DEC-001).

Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." NSM-DDR-001 had adopted D1 to D13 for TRL 3 work, open for his review, and left O1 to O4 as "Proposed, awaiting Amish". O2, O3 and O4 each carried a recommendation; O1 did not. TRL 4 remains on hold by Amish's instruction, and the repo stays at TRL 3.

Decision

Table 1. Items newly decided and what changed in the repo.

#	Item	Decision (the recommended option)	What changed
D1	Budget scope	`budget_usd` stays $100 and covers the NoiseMap parts only; FieldNode core costed in FieldNode	No change to `budget_usd` ($100); R13 already restated in NSM-REQ-001 v0.3; wording updated to "decided"
D2	Privacy claim	Guarantee rests on open head firmware, read-out protection and a published build hash; never "by physics"	Status wording only
D3	Microphone	Adapter board for ICS-43434 or IM72D128	Status wording only
D4	Processing location	Levels computed in the head	Status wording only
D5	Metrics and interval	15-minute, 22-byte records with 1-minute LAeq and statistics	Status wording only
D6	Wind flagging study	Both options studied at TRL 3	Superseded by O3
D7	Calibration	Shared 94 dB calibrator; one class 1 side-by-side per deployment	Status wording only; the side-by-side is TRL 4 work, on hold
D8	Public notice	Notice on each pole linking to the repository	Status wording only
D9 to D13	Precis choices	Levels only; standard FieldNode core; microphone at 4.0 m, 0.45 m off the pole; A and C weighting; printed ASA head	Status wording only
O2	R10 mass	Option (c): pocket the V-blocks and saddle, and relax R10 from 3 kg to 3.5 kg	`cad/src/model.py`: V-blocks pocketed from top and bottom (4 mm walls, 8 mm web), saddle pocketed 6 mm deep; the V notch geometry was also corrected (v0.1 had cut each block to a flat 10 mm plate). STEP and STL re-exported. NSM-DWG-001 Rev P1 to P2. NSM-REQ-001 R10 target 3 kg to 3.5 kg. NSM-CAL-001 v0.1 to v0.2: mass 3.32 kg (flawed geometry) to 3.45 kg pocketed, 3.96 kg solid; R10 not met to met on paper, margin 0.05 kg. BOM lines 7 and 14 describe the pockets; prices unchanged
O3	R11 wind flag method	Option (a): level-based flag from the Z-weighted band below 40 Hz; rain flagged on the server	No anemometer added. NSM-PRC-001 and NSM-REQ-001 state the method; NSM-CAL-001 records the anemometer as a rejected option. The field comparison that sets the threshold is TRL 4 work, on hold
O4			

#	Item	Decision (the recommended option)	What changed
	R12 airtime at SF10 and slower	Lengthen the interval automatically: 24 min at SF10, 48 min at SF11, 87 min at SF12	Firmware rule written into NSM-PRC-001 (how it works, step 5) and NSM-REQ-001 R12. NSM-CAL-001 [F2c]: at most 30.0 s/day at every spreading factor; R12 at risk to met on paper. Aligns with FieldNode's interval item (cross-repo action)

Table 2. Item left open here, decided on 2026-10-02.

#	Item	Status
O1	First co-design partner and street. No recommendation was made.	Decided by Amish, 2026-10-02 (recommendation approved, NSM-DEC-001): first partner to approach is a city environmental noise team or a university acoustics group that can put a Class 1 reference sound level meter beside the node on the test street; a Sensor.Community group is a useful second contact for volunteer hosts

Consequences

- Requirement status (NSM-CAL-001 v0.2): 0 not met, 3 at risk (R3, R4, R5), 4 not verifiable at TRL 3 (R8, R9, R11, R14), 4 met on paper (R7, R10, R12, R13), 4 met by design (R1, R2, R6, R15). Before: 1 not met (R10), 4 at risk (R3, R4, R5, R12).
- The R10 margin is only 0.05 kg; any added part needs a mass check. Shorter V-blocks or dropping FieldNode's unused small-pole V-blocks would add margin.
- NSM-PRB-001, NSM-PRC-001 and NSM-REQ-001 go from v0.3 to v0.4; NSM-DDR-001 from v0.1 to v0.2; NSM-CAL-001 from v0.1 to v0.2. Concept media are re-rendered from the updated model.
- Cross-repo actions, not made here: ask FieldNode for a street pole mounting variant for 60 to 140 mm poles; align the interval rule with FieldNode's interval item; agree the M12 pinout (FieldNode O2).
- TRL 4 is on hold by Amish's instruction. `trl` and `trl_target` stay at 3. Nothing here authorizes building, testing, purchasing, PCB work or firmware beyond a sketch.